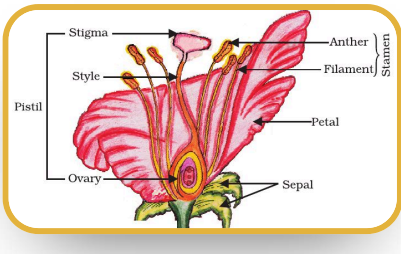
MT Classes, Motihari

How do Organisms Reproduce?



Class 10 Science, Chapter 7 (NCERT, CBSE 2026-27)

DNA copying and variation, fission to spores, flowers and seeds, and human reproduction, explained beside the textbook's own diagrams, with hands-on tools, real CBSE board questions with answers, every NCERT question solved and a quiz.

📌 Chapter snapshot

- Chapter 7 of NCERT Class 10 Science (reprint 2026-27), in Unit II “World of Living”, which carries **25 marks** in the 80-mark theory paper together with Life Processes, Control and Coordination, and Heredity.
- Most-asked in boards: **diagrams** (budding in Hydra, spore formation in Rhizopus, L.S. of flower, germination of pollen on stigma), **pollination vs fertilisation**, **DNA stability through meiosis**, and **contraception**.
- Want a shorter version? See our quick revision notes for this chapter.

Start

Why do organisms reproduce?

Reproduction is the process by which living organisms produce new individuals of their own kind.

- Nutrition, respiration and excretion are needed to keep an individual **alive**. Reproduction is **not**: an individual can live its whole life without reproducing.
- Reproduction even costs the parent a lot of **energy**. So why do it? Because it keeps the **species** going after the individuals die.
- We notice a kind of organism only because there are **large numbers** of it. Those numbers come from reproduction.
- We say two organisms belong to the same species because they **look similar**: reproduction makes new individuals that look very much like the parents.



Chapter opener: pollen tube growing to the ovule

⭐ **Exam line**

Reproduction is **not essential for the survival of an individual**, but it is **essential for the survival (continuity) of a species**.

7.1

Do organisms create exact copies of themselves?

DNA: the blueprint of body design

- Organisms look similar because their **body designs** are similar, and body designs are similar because their **blueprints** are similar.
- The blueprint is the **DNA (Deoxyribonucleic Acid)** in the chromosomes of the nucleus. DNA is the information source for making **proteins**. Changed information → different proteins → altered body design.
- So the **basic event in reproduction is the creation of a DNA copy**.

How a cell makes a copy of itself

1. The cell uses **chemical reactions** to build a copy of its DNA. Now there are two copies.
2. Simply pushing one copy out would not work: it would have no cell structure to keep life going.
3. So DNA copying goes along with making **additional cellular apparatus**.
4. The two DNA copies separate, each with its own cellular apparatus: **one cell divides into two**.

Are the two cells identical?

No bio-chemical reaction is absolutely reliable, so DNA copying makes small **variations** each time. The copies are **similar but not identical**.

Drastic variation	Small variation
The new DNA cannot work with the inherited cell apparatus The newborn cell simply dies.	**The cell survives** Surviving cells are similar to, but subtly different from, each other. This inbuilt tendency for variation is the **basis of evolution** (next chapter).

7.1.1 The importance of variation

- Populations fill well-defined places called **niches** in the ecosystem. **Consistent** DNA copying keeps the body design that suits that niche, so reproduction is linked to the **stability of populations**.
- But niches can change for reasons beyond the organism's control: temperature goes up or down, water levels change, meteorites hit.
- If everyone were identical, a drastic change could wipe out the whole population. If a few individuals carry **variations**, some may survive.
- **NCERT example:** bacteria in temperate water. If global warming heats the water, most die, but the few **heat-resistant variants** survive and multiply. Variation is useful for the **survival of species over time**, though not necessarily for every individual.

7.2 Modes of reproduction used by single organisms

The mode an organism uses **depends on its body design**. All the modes below create a new generation from a **single individual**: this is **asexual reproduction**.

Mode	Who uses it	One-line idea
Binary fission	Amoeba, Leishmania, many bacteria and protozoa	One cell splits into two equal halves.
Multiple fission	Plasmodium (malarial parasite)	One cell divides into many daughter cells at the same time.
Budding	Yeast, Hydra	A small outgrowth (bud) forms, grows and separates.
Fragmentation	Spirogyra	The body breaks into pieces on maturing; each piece grows.
Regeneration	Hydra, Planaria	A cut piece grows into a complete organism.
Vegetative propagation	Bryophyllum, potato, money-plant, sugarcane, rose, grapes	Root, stem or leaf grows into a new plant.
Spore formation	Rhizopus (bread mould)	Sporangia release thick-walled spores.

Activities 7.1 and 7.2: yeast and bread mould

– **Activity 7.1: Yeast in sugar solution, observed after 1–2 hours. What do you see?**

Yeast cells with **small buds** attached, some buds with buds of their own (chains). Yeast reproduces by **budding**; the sugar gives it food and the warm place speeds it up.

– **Activity 7.2: Wet bread kept in a cool, moist, dark place for a week. What do you see?**

White cottony threads first, then tiny black dots. The threads are **hyphae** of bread mould (*Rhizopus*); the black "blob-on-a-stick" heads are **sporangia** full of spores. Mould grows by **spore formation**, while yeast grows by budding. Moisture is essential: dry bread does not grow mould.

7.2.1 Fission

In **unicellular** organisms, cell division itself (**fission**) creates new individuals.

Binary fission

- Many bacteria and protozoa simply **split into two equal halves**.
- In ***Amoeba***, the splitting can take place in **any plane** (Fig. 7.1 a). First the nucleus divides, then the cytoplasm.
- ***Leishmania*** (causes **kala-azar**) has a whip-like **flagellum** at one end, so binary fission happens in a **definite orientation** (lengthwise) in relation to this structure (Fig. 7.1 b).

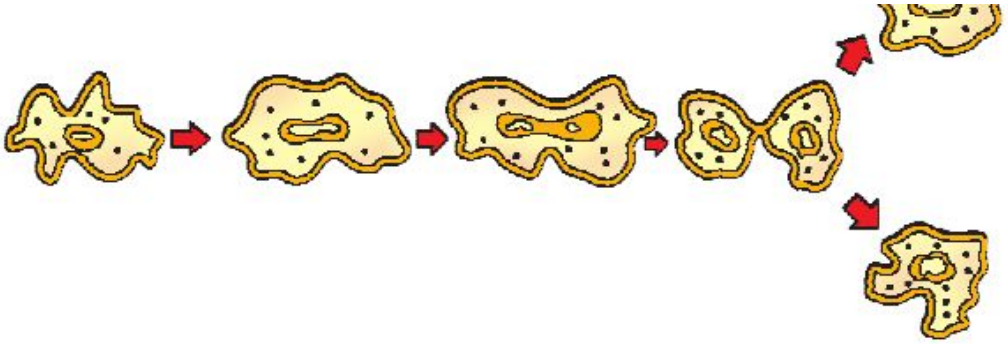


Fig. 7.1 (a) Binary fission in *Amoeba*

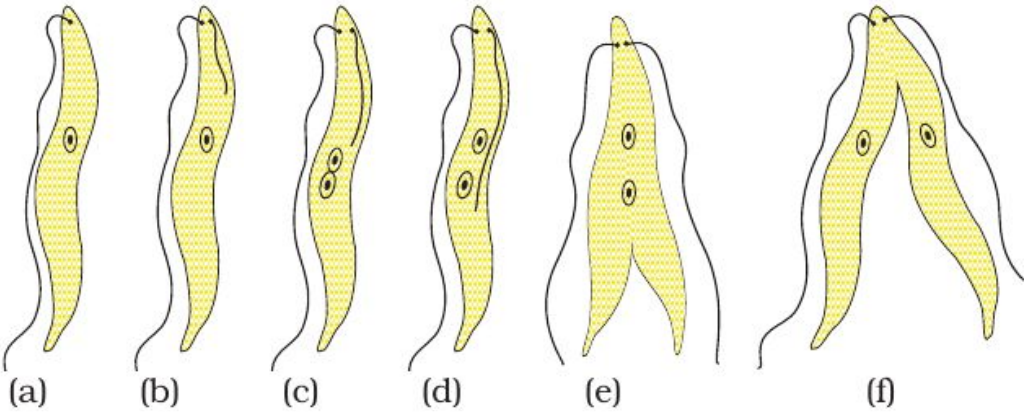


Fig. 7.1 (b) Binary fission in *Leishmania*

Multiple fission

- The malarial parasite ***Plasmodium*** divides into **many daughter cells simultaneously**.
- The nucleus divides many times, each nucleus gets some cytoplasm, and when the outer wall breaks, many daughter cells are released (Fig. 7.2).
- Often happens when conditions are unfavourable; a protective cover (cyst) forms around the cell.

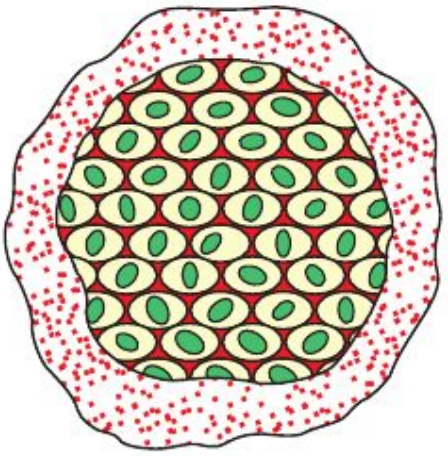


Fig. 7.2 Multiple fission in *Plasmodium*

– **Activity 7.3: Compare a permanent slide of Amoeba with one showing binary fission.**

The normal Amoeba has one nucleus and an irregular shape. The dividing Amoeba shows a **lengthened or divided nucleus** and a **constriction** in the cytoplasm, producing two daughter cells.

7.2.2 Fragmentation

- Multicellular organisms with a **simple body organisation** can still use simple methods. ***Spirogyra*** (a green filamentous alga) simply **breaks into smaller pieces** on maturation; each **fragment** grows into a new individual.
- Why does this work? Activity 7.4 shows that Spirogyra filaments have **no different tissues**: every cell is alike and can do everything.
- Most multicellular organisms cannot divide cell-by-cell, because specialised cells form **tissues**, tissues form **organs**, and organs must be at definite positions. They need more complex ways, with reproduction done by a **specific cell type** that can grow, multiply and make other cell types.

– **Activity 7.4: Can you identify different tissues in the Spirogyra filaments?**

No. All cells of the filament look the same (each with a ribbon-like spiral chloroplast). There are **no specialised tissues**, which is why any fragment can grow into a new filament.

7.2.3 Regeneration and budding

Regeneration

- Many fully differentiated organisms can give rise to new individuals from **body parts**. ***Hydra*** and ***Planaria*** can be cut into any number of pieces and each piece grows into a complete organism (Fig. 7.3).
- It is done by **specialised cells** that multiply into a mass of cells; these then change into different cell types and tissues in an organised sequence called **development**.
- **Regeneration is not the same as reproduction**: most organisms do not normally depend on being cut up to reproduce.

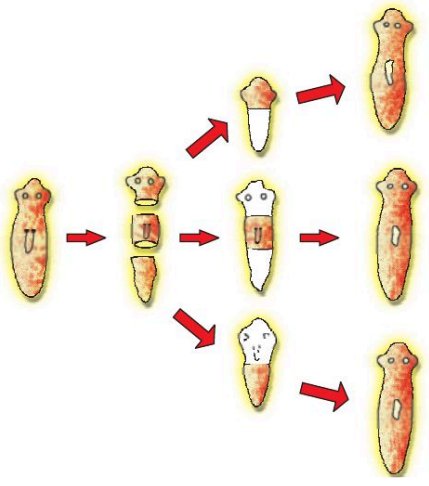


Fig. 7.3 Regeneration in *Planaria*

7.2.4 Budding

Hydra uses its **regenerative cells** for reproduction by budding. A **bud** develops as an outgrowth due to **repeated cell division at one specific site**. It develops into a tiny individual and, when fully mature, **detaches** from the parent and lives independently. **Yeast** also reproduces by budding.

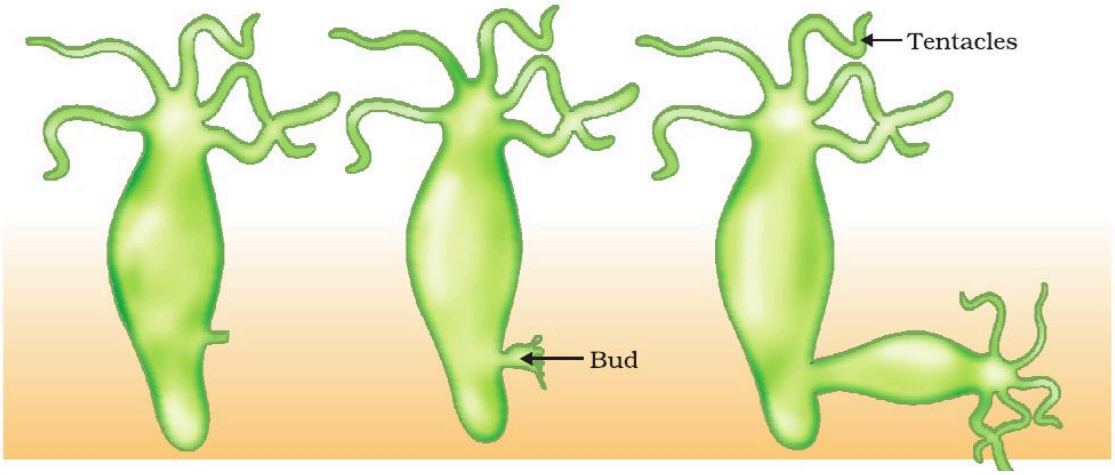


Fig. 7.4 Budding in *Hydra*

> ✍️ **Diagram tip (asked in 2024 and 2025 boards)**
>
> Draw at least **three stages**: parent with a small bud, bud with tentacles, bud detaching. Label **tentacles**, **bud** and **parent Hydra**.

7.2.5 Vegetative propagation

In many plants, **root, stem or leaves** develop into new plants under suitable conditions. This is **vegetative propagation**. Methods like **layering** and **grafting** use it to grow sugarcane, roses and grapes.

- ***Bryophyllum***: buds in the **notches along the leaf margin** fall on soil and grow into new plants (Fig. 7.5).
- **Potato**: the "eyes" (notches) on the tuber are buds that sprout.
- **Money-plant**: a stem piece with a node (leaf) grows roots.

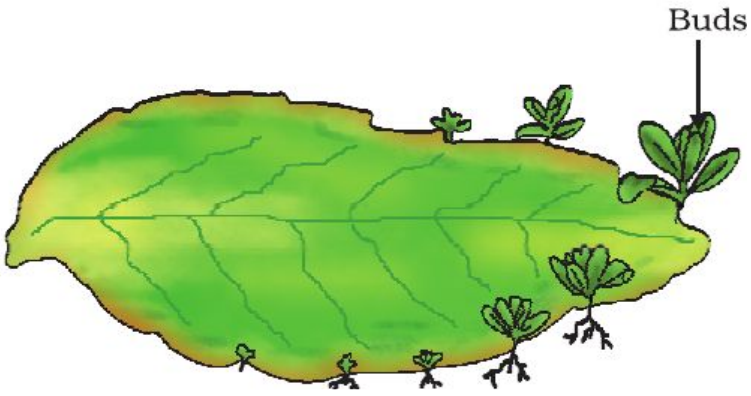


Fig. 7.5 Leaf of *Bryophyllum* with buds

Advantages of vegetative propagation (learn all four)

1. Plants **bear flowers and fruits earlier** than those grown from seeds.
2. Plants that have **lost the capacity to produce seeds** (banana, orange, rose, jasmine) can be grown.
3. All plants are **genetically similar** to the parent, so they keep all its desirable characters.
4. It is quick, cheap and easy, e.g. sugarcane, rose, grapes.

– **Activity 7.5: Which potato pieces give rise to fresh green shoots and roots?**

Only the pieces **with a notch (bud / eye)**. Pieces without a bud just rot or dry. The buds are the growing points of the stem.

– **Activity 7.6: Money-plant pieces with a leaf vs pieces between two leaves. Which grow?**

Pieces **with at least one leaf (a node)** grow roots and fresh leaves. Pieces from **between two leaves (internode)** do not. Conclusion: new plants come from the **buds at the nodes**.

More to know: tissue culture

1. Remove tissue or cells from the **growing tip** of a plant.
2. Place them in an **artificial medium**; they divide rapidly into a small group of cells called **callus**.
3. Move the callus to a medium with **hormones** for growth and differentiation; plantlets form.
4. Plant the plantlets in soil to grow into mature plants.

Many plants can be grown from one parent in **disease-free** conditions. Commonly used for **ornamental plants**.

7.2.6 Spore formation

- In bread mould (***Rhizopus***), the thread-like structures are **hyphae**: they are **not** reproductive.
- The tiny **blob-on-a-stick** structures are reproductive: the blobs are **sporangia**, which contain **spores**.
- Each spore can develop into a new Rhizopus. Spores have **thick walls** that protect them until they reach another **moist surface** and can begin to grow.

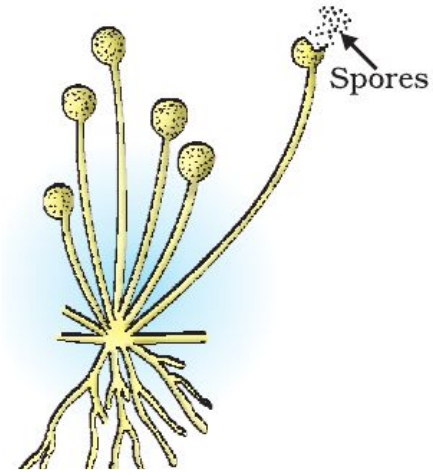


Fig. 7.6 Spore formation in *Rhizopus*

> ✍️ **Diagram tip (asked in 2024 boards)**
>
> Label **sporangium** and **spores** as the reproductive parts, and **hyphae** as the non-reproductive part.

7.3 Sexual reproduction

Bulls alone cannot produce calves, nor hens alone chicks. When **two individuals (male and female)** are needed to produce a new generation, it is **sexual reproduction**.

7.3.1 Why the sexual mode of reproduction?

1. Errors in DNA copying cause **variations**, which help a **population** survive. So it makes sense to generate more variation.
2. But DNA copying is quite accurate, so variation builds up **slowly**. (Making it less accurate would kill many cells.)
3. Each individual already carries variations **accumulated over generations**, all proven harmless because the individual is alive.
4. **Combining DNA from two individuals** creates new combinations of variations quickly. That is what sexual reproduction does.

The problem of doubling DNA, and its solution

- If each new generation combined the full DNA of two parents, the DNA would **double every generation** and mess up the control of the cellular apparatus.

- Solution: special cell lineages in **specialised organs** make **germ-cells (gametes)** with **half the chromosomes and half the DNA**, by a cell division called **meiosis**.
- When two gametes fuse, the **original number is re-established** in the zygote. This keeps the DNA content of a species **stable**.

Two kinds of gametes

- A zygote that grows into a complex organism needs **stored energy (food)**.
- In very simple organisms the two germ-cells are alike. In complex ones they specialise: one is **large with food stores** (the **female gamete**, egg), the other is **small and motile** (the **male gamete**, sperm).
- Making these two types leads to differences in male and female reproductive organs and, in some cases, bodies.

7.3.2 Sexual reproduction in flowering plants

The reproductive parts of **angiosperms** are in the **flower**: sepals, petals, stamens and pistil. **Stamens and pistil** are the reproductive parts that contain the germ-cells. (Sepals protect the bud; petals attract pollinators.)

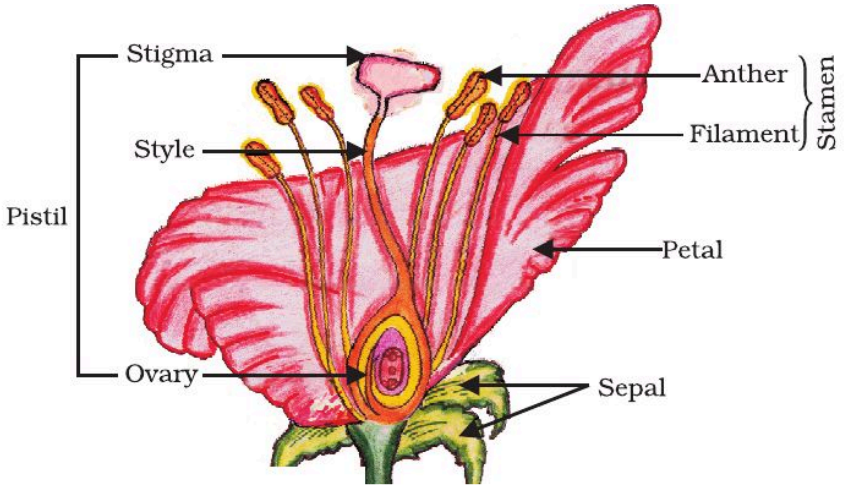


Fig. 7.7 Longitudinal section of flower

Stamen (male) = **anther** + **filament**. The anther makes yellowish **pollen grains**.

Pistil (female, centre) = **stigma** (sticky tip) + **style** (middle tube) + **ovary** (swollen base).

The ovary contains **ovules**; each ovule has an **egg cell**.

Unisexual flower	Bisexual flower
Has **either** stamens **or** pistil	Has **both** stamens and pistil
Papaya, watermelon	Hibiscus, mustard

Pollination

Transfer of pollen from the **anther to the stigma** is **pollination**.

Self-pollination

Same flower

Pollen reaches the stigma of the same flower (or another flower on the same plant).

No agent needed

Possible only in bisexual flowers.

Cross-pollination

Another flower

Pollen goes to a flower of another plant of the same species.

Needs an agent

Wind, water or animals (insects, birds). Brings variation.

Fertilisation

1. Pollen lands on a **suitable stigma**.
2. A **pollen tube** grows out of the pollen grain and travels **through the style** to the ovary.
3. The **male germ-cell** moves down the tube into the ovule.
4. It fuses with the **female germ-cell (egg)**: this fusion is **fertilisation**, giving a **zygote**.

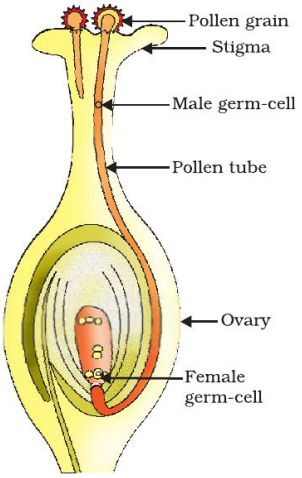


Fig. 7.8 Germination of pollen on stigma

✍ Asked in 2025 and 2026 boards

Draw Fig. 7.8 and label **pollen grain, stigma, pollen tube, male germ-cell, ovary, female germ-cell**. In 2025 the same figure came with parts marked A, B, C to identify.

After fertilisation

Part	Becomes
Zygote	divides many times → **embryo** (inside the ovule)
Ovule	develops a tough coat → **seed**
Ovary	grows rapidly and ripens → **fruit**

Part	Becomes
Petals, sepals, stamens, style, stigma	may shrivel and fall off (sometimes the sepals persist, e.g. on brinjal and tomato)

Germination

The seed contains the future plant, the **embryo**, which develops into a seedling under suitable conditions (water, air, warmth). This is **germination**.

- **Plumule**: future shoot.
- **Radicle**: future root.
- **Cotyledon**: food store for the growing embryo.

Advantages of seeds: a protective coat, stored food, they can wait (stay dormant) until conditions are right, and they can be dispersed far away.

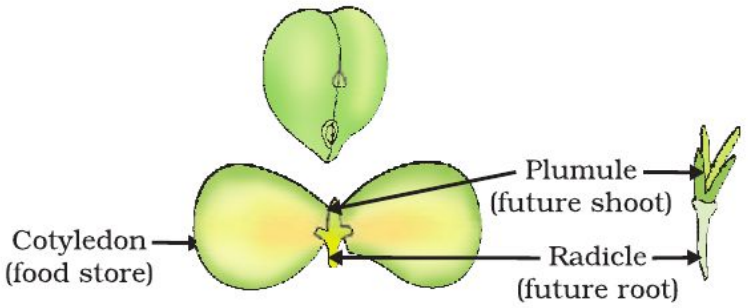


Fig. 7.9 Germination

– **Activity 7.7: Soaked Bengal gram (chana) seeds cut open. Which parts do you see?**

Two big **cotyledons** (food store), and between them a tiny embryo axis: the **plumule** (future shoot) and the **radicle** (future root). The seed coat peels off easily after soaking.

7.3.3

Reproduction in human beings

Humans use the sexual mode. As we grow, the body first just gets bigger (growth). In the early teenage years, a new set of changes appears that changes how the body looks: this is **sexual maturation**, and the period is **puberty**.

Changes in both boys and girls

Thick hair

In armpits and the genital area; thinner hair on legs, arms and face.

Oily skin, pimples

Different in boys and girls

Girls

Breast size increases, skin of the nipples darkens, menstruation begins.

Boys

> Becoming aware of one's own and others' bodies in new ways.

> Thick facial hair, voice begins to crack, the penis occasionally becomes enlarged and erect.

- These changes are **slow** (months and years), do not happen at an exact age, and differ from person to person.
- **Why at this age?** While the body is growing, resources go mainly to growth. As general growth slows, the **reproductive tissues mature**.
- New hair patterns etc. are **signals** that sexual maturation is happening. Special organs are needed to transfer germ-cells, and in mammals the female body and breasts mature to carry and later breast-feed a baby.

7.3.3 (a) Male reproductive system

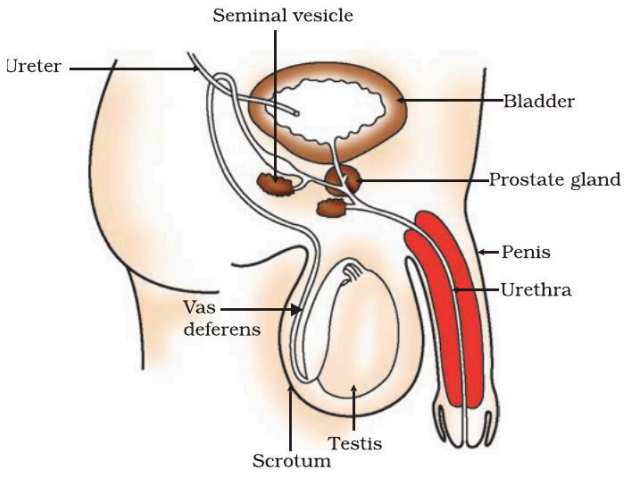


Fig. 7.10 Human male reproductive system

1. **Testes**: make **sperms** and the hormone **testosterone**. They lie **outside the abdomen in the scrotum** because sperm formation needs a **lower temperature** than body temperature.
2. **Vas deferens**: carries sperms; unites with the tube from the urinary bladder.
3. **Urethra**: **common passage** for sperms and urine.
4. **Seminal vesicles and prostate gland**: add secretions so sperms are in a **fluid** that eases transport and gives **nutrition**.
5. **Penis**: organ for transferring sperms.

Testosterone regulates sperm formation and brings the changes seen in boys at puberty. **Sperms** are tiny bodies of mainly genetic material with a long **tail** to swim towards the egg.

7.3.3 (b) Female reproductive system

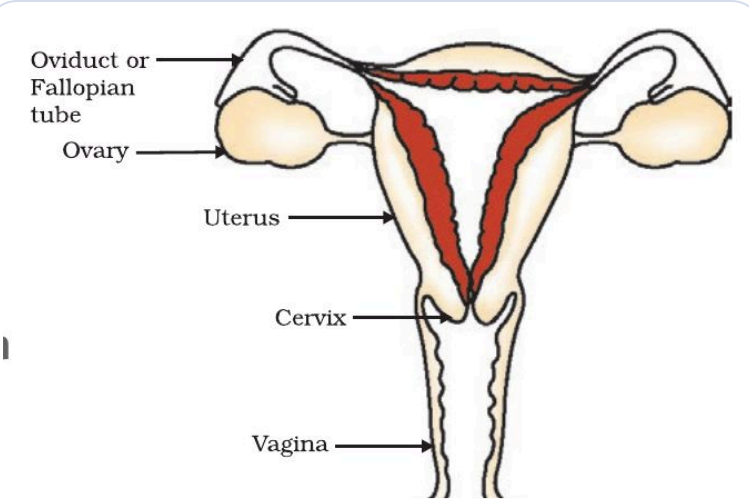


Fig. 7.11 Human female reproductive system

1. **Ovaries**: make **eggs** and some **hormones** (oestrogen, progesterone). At birth they already hold thousands of immature eggs; from puberty **one egg is released every month** by one of the ovaries.
2. **Oviduct (fallopian tube)**: carries the egg to the uterus; **site of fertilisation**.
3. **Uterus**: elastic bag where the two oviducts unite; the embryo implants and grows here.
4. **Cervix**: the uterus opens into the vagina through it.
5. **Vagina**: receives sperms; birth canal.

Fertilisation and development

1. Sperms enter through the vagina during intercourse and travel up to the **oviduct**, where they may meet the egg.
2. The fertilised egg (**zygote**) divides into a ball of cells, the **embryo**.
3. The embryo is **implanted** in the thick, blood-rich lining of the **uterus** and develops organs to become a **foetus**.
4. Nutrition comes through the **placenta**: a disc embedded in the uterine wall, with **villi** on the embryo's side and **blood spaces** on the mother's side. This gives a large surface area for **glucose and oxygen** to pass to the embryo, and for **wastes** to pass back into the mother's blood.
5. Development takes about **nine months**. The child is born by **rhythmic contractions of the uterus muscles**.

7.3.3 (c) What happens when the egg is not fertilised?

The egg lives for about **one day**. Every month the uterus lining becomes **thick and spongy** to receive a fertilised egg. If there is no fertilisation, the lining is not needed: it slowly breaks down and comes out through the vagina as **blood and mucous**. This monthly cycle is **menstruation**, lasting about **2 to 8 days**.

7.3.3 (d) Reproductive health

- Some sexual maturation does **not** mean the body or mind is ready for sexual acts or for having and bringing up children.
- There can be pressure from friends, from family (to marry and have children) or from government agencies (to avoid children). Making choices can be difficult.

Sexually transmitted diseases (STDs)

Bacterial	Viral
Gonorrhoea, syphilis	Warts, HIV-AIDS

A **condom** on the penis during sex helps prevent the transmission of many of these infections to some extent.

Contraception: methods to avoid pregnancy

Pregnancy makes major demands on a woman's body and mind; if she is not ready, her health suffers. Contraceptive methods fall into categories:

Category	Examples	How it works	Drawback
Mechanical barrier	Condom; similar covering in the vagina	Sperm cannot reach the egg	Few; also protects against STDs
Hormonal (chemical)	Oral pills	Change hormonal balance so eggs are not released	Side-effects, as hormone balance changes
Intra-uterine devices	Loop, Copper-T	Placed in the uterus to prevent pregnancy	Side-effects from irritation of the uterus; **no protection from STDs**
Surgical	Vasectomy (vas deferens blocked), tubectomy (fallopian tube blocked)	Sperm transfer / egg movement is blocked	Safe in the long run, but surgery can cause infection if not done properly

- Surgery can also remove unwanted pregnancies, but it is **misused** for illegal **sex-selective abortion of female foetuses**. Prenatal sex determination is **prohibited by law**. For a healthy society the female-male **sex ratio** must be maintained.

- Birth and death rates decide population size. A growing population makes it harder to raise living standards, but if **inequality** is the main reason for poor living standards, population size is relatively less important.



Differences to learn (3-mark favourites)

Asexual vs sexual reproduction

Asexual	Sexual
One parent	Two parents (male and female)
No gametes, no fertilisation	Gametes form (by meiosis) and fuse
Offspring nearly identical to the parent	Offspring show more variation
Fast; many offspring	Slower
Amoeba, Hydra, Bryophyllum	Flowering plants, humans

Binary vs multiple fission

Binary fission	Multiple fission
Parent divides into **two** daughter cells	Parent divides into **many** daughter cells at once
Nucleus divides once	Nucleus divides many times
Usually in favourable conditions	Often in unfavourable conditions, inside a cyst
Amoeba, Leishmania	Plasmodium

Pollination vs fertilisation (asked in 2026)

Pollination	Fertilisation
Transfer of pollen from anther to stigma	Fusion of male and female germ-cells
Happens first	Happens after pollination
Outside the ovary (on the stigma)	Inside the ovule, in the ovary
Needs agents like wind, water, animals (for cross)	Needs a pollen tube to carry the male germ-cell
Result: pollen on stigma	Result: zygote

Self vs cross-pollination (asked in 2023 and 2025)

Self-pollination	Cross-pollination
Pollen to stigma of the same flower or another flower of the same plant	Pollen to stigma of a flower on a different plant (same species)
No external agent needed	Needs wind, water or animals
Little variation	Brings genetic variation

Memory tricks

Amoeba vs Leishmania any vs definite

Amoeba splits in **A**ny plane; **L**eishmania splits **L**engthwise (definite plane) because of its flagellum.

Plasmodium P = Plenty

Plasmodium makes **P**lenty of daughters at once: multiple fission.

Hydra does two H = 2

Hydra: budding and regeneration. Planaria: regeneration. Yeast: budding.

Pistil top to bottom S S O

Stigma, **S**tyle, **O**vary: "**S**ticky **S**traw to the **O**ven".

What becomes what big → big

Bigger word **ovary** → bigger thing, **fruit**. Smaller **ovule** → smaller **seed**.

Male glands SPerm helpers

Seminal vesicles + **P**rostate add fluid: easier transport + nutrition.

Copper-T and STDs T ≠ protecTion

Only **condoms** (barriers) give some STD protection. Copper-T, pills and surgery do not.



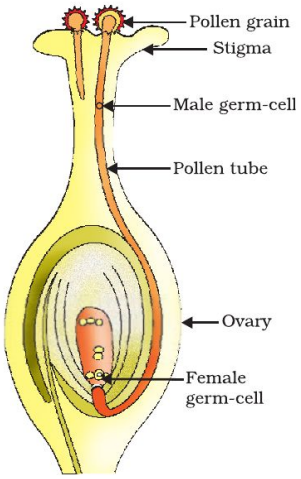
Previous year board questions

These questions come from the CBSE Class 10 Science board papers of the years shown, with the paper code (set) and marks. Tap a question to see a model answer.

Where a question had options or a diagram we could not check word for word, it is **reworded** and marked so. The years and marks shown are only those we could confirm from the paper or the official marking scheme.

– **2026 · 31/1/1 · 2 marks**

Draw a neat diagram to show germination of pollen on the female reproductive part of the flower.



Model diagram: label pollen grain, stigma, pollen tube, male germ-cell, ovary, female germ-cell

Draw the pistil with pollen grains on the stigma and one pollen tube growing down the style into the ovule. Labels: **pollen grain, stigma, pollen tube, male germ-cell, style, ovary, female germ-cell**.

– **2026 · 31/1/1 · 2 marks (OR)** **List any two differences between pollination and fertilisation.**

(i) **Pollination** is the transfer of pollen from anther to stigma; **fertilisation** is the fusion of the male germ-cell with the female germ-cell (egg). (ii) Pollination happens on the stigma, outside the ovary, and comes first; fertilisation happens inside the ovule in the ovary, after the pollen tube reaches it. (Also: pollination may need agents like wind or insects; fertilisation produces a zygote.)

– **2026 · 31/1/1 · 1 mark (reworded)**

Identify the correct statement about reproduction in Spirogyra, Leishmania and Hydra.

The options are not reproduced here; know these facts: **Spirogyra** reproduces by **fragmentation**, **Leishmania** by **binary fission** (in a definite plane), and **Hydra** by **budding** (and can also regenerate). All three are **asexual**.

2026 · 31/1/1, Q16 (also in set 31/1/3), 1 mark each: “Given below is a certain situation. Analyse it and describe its possible impact.”

– 2026 · 1 mark **The sperm encounters the egg when it reaches the oviduct in human females.**

Fertilisation may take place in the oviduct, forming a zygote, which divides into an embryo and gets implanted in the uterus lining: pregnancy begins.

– 2026 · 1 mark **Self-pollination does not occur in a flower that contains only pistil.**

The flower is **unisexual (female)** with no stamens, so it has no pollen of its own. It can be pollinated only by **cross-pollination**, through agents like wind, water or insects, which also brings variation.

– 2026 · 1 mark **Egg does not get fertilised in a human female.**

The egg lives about one day and dies. The thick, spongy uterus lining is not needed, so it breaks down and comes out through the vagina as blood and mucous: **menstruation** (2–8 days).

– 2026 · 1 mark **The seed is placed under appropriate conditions of water and air in the soil.**

The seed **germinates**: the embryo grows, the **radicle** forms the root and the **plumule** the shoot, using food stored in the cotyledons, and a seedling develops.

– 2026 · 1 mark (OR) **Spores are liberated from the blob-like structures of the bread mould.**

The spores (from the sporangia) spread; protected by thick walls, each one that lands on a **moist surface** germinates and grows into a new *Rhizopus*. The mould spreads quickly.

– 2026 · 1 mark (OR) **What happens when a Bryophyllum leaf falls on wet soil?**

The **buds in the notches** along the leaf margin develop roots and shoots and grow into **new Bryophyllum plants** (vegetative propagation).

– 2026 · 1 mark (OR)

A pollen grain from a different species lands on the stigma of a totally unrelated species.

It is not a **suitable stigma**, so the pollen does not germinate (no pollen tube grows) and **no fertilisation** takes place; no seed forms from it.

– 2026 · 1 mark (OR) **Copper-T is placed in the uterus of a human female.**

It **prevents pregnancy**, but it may cause side effects due to irritation of the uterus, and it gives **no protection against STDs**.

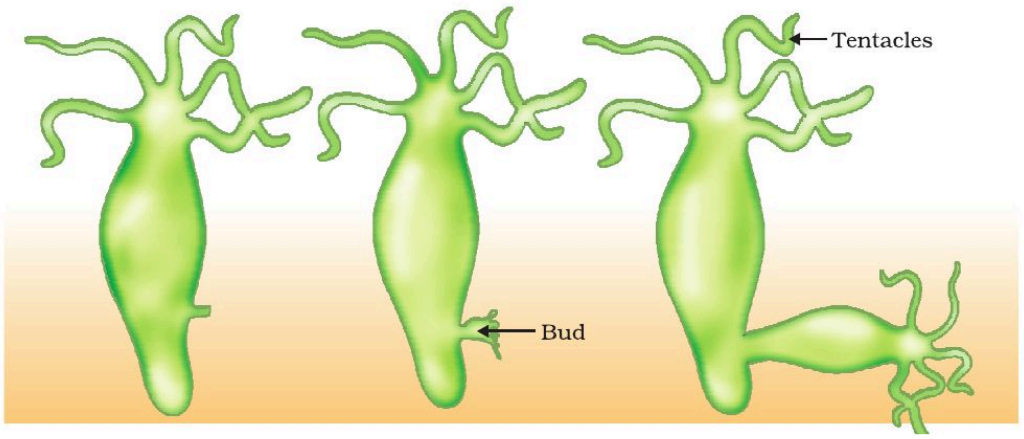
– 2026 · 1 mark (OR) **Spirogyra breaks into smaller fragments upon maturation.**

Each fragment grows into a **new Spirogyra filament** (fragmentation), because all its cells are alike and each can divide.

– 2025 · 31/1/1 · 1 mark (reworded) **Which statements are true for a bisexual flower?**

A bisexual flower **possesses both stamens and pistil** (male and female parts), so it **can** self-pollinate. Examples: Hibiscus, mustard. (Papaya and watermelon have unisexual flowers.)

– 2025 · 31/1/1 · 2 marks **Draw labelled diagrams to show different stages of budding in Hydra.**



Model answer: draw these three stages

Stage 1: a small **bud** arises as an outgrowth on the parent's body due to repeated cell division at one site. Stage 2: the bud grows and develops a mouth and **tentacles**. Stage 3: the mature bud **detaches** and becomes an independent Hydra. Label: parent Hydra, bud, tentacles.

– 2025 · 31/1/1 · 3 marks

(a) What are chromosomes? (b) Explain in brief how stability of DNA content of a species is ensured in sexually reproducing organisms.

(a) **Chromosomes** are thread-like structures in the nucleus that carry hereditary information in the form of **DNA**. (b) Specialised cells in the reproductive organs divide by **meiosis**, so each **gamete** gets **half** the number of chromosomes and half the DNA. When a male and a female gamete fuse at fertilisation, the **zygote** gets back the **full number**. So the chromosome number and DNA content stay the same from generation to generation.

– 2025 · 31/1/1 · 5 marks

(i) Write the functions of the following parts of the human female reproductive system: (I) Ovary (II) Fallopian tube (III) Uterus. (ii) State briefly two contraceptive methods used by human males.

(i) **Ovary**: produces eggs (one each month) and female hormones such as oestrogen. **Fallopian tube**: carries the egg from the ovary to the uterus and is the **site of fertilisation**. **Uterus**: the embryo is implanted in its lining and develops into a foetus; it is nourished through the placenta; its muscles contract at birth.
(ii) **Condom**: a mechanical barrier on the penis, so sperms do not reach the egg (also gives some protection from STDs). **Vasectomy**: the vas deferens is blocked by surgery, so sperms cannot pass through.

– 2025 · 31/1/1 · 5 marks (OR)

(i) Differentiate between self-pollination and cross-pollination. (ii) Identify A, B and C in the given diagram (pollen germinating on the stigma) and write one function of each.

(i) Self-pollination: pollen from the anther reaches the stigma of the **same flower** (or same plant); no agent needed. Cross-pollination: pollen reaches the stigma of a flower on **another plant**; needs wind, water or animals; brings variation.
(ii) Based on Fig. 7.8: **A = stigma**: receives pollen and lets it germinate. **B = pollen tube**: carries the male germ-cell through the style to the ovule. **C = female germ-cell (egg)**: fuses with the male germ-cell to form the zygote.

– 2024 · 31/1/1 · 1 mark **In which organism is multiple fission a means of asexual reproduction?**

Plasmodium (the malarial parasite): one cell divides into many daughter cells at the same time.

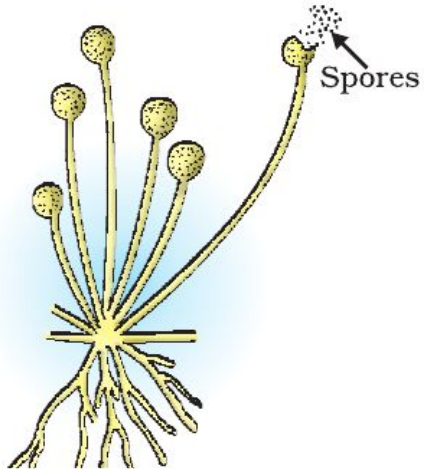
2024 · 31/1/1 · 5 marks

(i) Name three techniques/devices used by human females to avoid pregnancy. (ii) What will happen if in a human female (a) fertilisation takes place, (b) an egg is not fertilised?

(i) **Oral pills** (hormonal): stop eggs from being released; can have side-effects because hormone balance changes. **Copper-T / loop** placed in the uterus: can irritate the uterus. **Surgical blocking of the fallopian tubes** (tubectomy): the egg cannot reach the uterus; surgery can cause infection if not done properly. (A vaginal covering, a barrier, is also accepted.)
(ii) (a) The zygote forms an embryo that is implanted in the uterus: **pregnancy**, with nourishment through the placenta. (b) The thick uterine lining breaks down and comes out as blood and mucous: **menstruation**.

2024 · 31/1/1 · 5 marks (OR)

(i) Draw a diagram showing spore formation in Rhizopus and label the (a) reproductive and (b) non-reproductive parts. Why does Rhizopus not grow on dry bread? (ii) Name and explain the process by which reproduction takes place in Hydra.



Model diagram: Rhizopus

(i) (a) Reproductive parts: **sporangia** containing **spores**. (b) Non-reproductive part: **hyphae** (thread-like). Rhizopus does not grow on dry bread because spores need **moisture** to germinate; they grow only on a moist surface.
(ii) **Budding**: a bud forms as an outgrowth by repeated cell division at one specific site (using regenerative cells), grows into a tiny Hydra with tentacles, and when mature detaches to live independently.

2023 · 31/1/1 · 5 marks (reworded)

(a) Name the non-reproductive parts of a flower. (b) Distinguish between self-pollination and cross-pollination, and give two points of significance of pollination. (c) What do the ovule and the ovary develop into after fertilisation?

(a) **Sepals** (calyx) and **petals** (corolla).
(b) **Self-pollination**: transfer of pollen from anther to stigma in the same flower or another flower of the same plant. **Cross-pollination**: transfer of pollen from the anther of one flower to the stigma of a flower on a different plant. **Significance** (any two): necessary for fertilisation and seed formation; stimulates fruit development; cross-pollination brings genetic variation.
(c) **Ovule → seed**; **ovary → fruit**.

Practice

Board-pattern practice

Extra questions in the current CBSE pattern (not from a specific year). Tap to check.

Assertion–Reason (1 mark each)

(a) Both A and R are true, and R explains A. (b) Both true, but R does not explain A. (c) A true, R false. (d) A false, R true.

1 **A: Testes are located outside the abdominal cavity in the scrotum. R: Sperm formation requires a lower temperature than normal body temperature.**

(a)

2 **A: Spirogyra reproduces by fragmentation. R: Spirogyra has a simple body without specialised tissues.**

(a)

3 **A: Regeneration is a common mode of reproduction in most animals. R: Planaria can regenerate from small pieces.**

(d) Most organisms do not depend on being cut up to reproduce, so A is false.

4 **A: Plants raised by vegetative propagation bear flowers and fruits earlier. R: They are produced from seeds.**

(c) They are produced from root, stem or leaf, not seeds.

5 **A: The urethra is a common passage for urine and sperms. R: The vas deferens unites with a tube coming from the urinary bladder.**

(a)

6 **A: Copper-T protects against sexually transmitted diseases. R: Copper-T is placed in the uterus to prevent pregnancy.**

(d)

Case-based (4 marks)

C1 **A student kept a moist slice of bread in a dark cupboard. After a few days, white threads with black dots appeared. (i) Name the organism. (ii) Name its mode of reproduction. (iii) What are the black dots? (iv) Why did the same experiment fail with dry bread?**

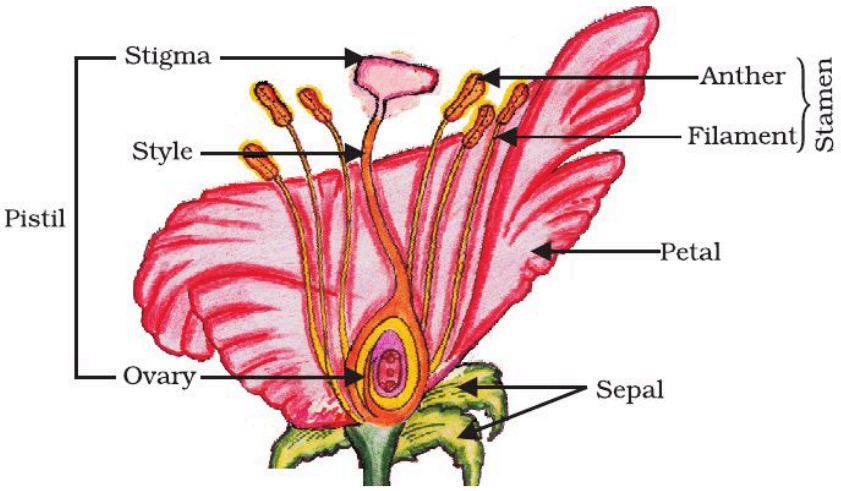
(i) *Rhizopus* (bread mould). (ii) Spore formation. (iii) Sporangia containing spores. (iv) Spores need moisture to germinate.

C2 **The placenta is a disc embedded in the uterine wall with villi on the embryo's side and blood spaces on the mother's side. (i) What is the role of the villi? (ii) Name two substances passed from mother to embryo. (iii) How are the embryo's wastes removed? (iv) How long does development take?**

(i) They give a large surface area for exchange. (ii) Glucose and oxygen. (iii) They pass into the mother's blood through the placenta. (iv) About nine months.

Long answer (5 marks)

– **L1** **Draw a labelled diagram of the longitudinal section of a flower and explain how a seed and fruit are formed after pollination.**



Model diagram: Fig. 7.7

After pollen lands on the stigma, a pollen tube grows through the style to the ovule and the male germ-cell fuses with the egg (fertilisation) to form a zygote. The zygote divides to form the embryo; the ovule develops a tough coat and becomes the **seed**; the ovary ripens into the **fruit**; petals, sepals, stamens, style and stigma shrivel and fall.

Recap What you have learnt

- Reproduction is **not essential** to keep an individual alive; it keeps the **species** going.
- Reproduction involves making a **DNA copy** and additional **cellular apparatus**. Copying is not perfect, so **variations** arise; they help species survive changes in their niche.
- The mode depends on **body design**. Fission (binary: Amoeba, Leishmania; multiple: Plasmodium), fragmentation (Spirogyra), regeneration (Planaria, Hydra), budding (Hydra, yeast), vegetative propagation (Bryophyllum, potato), spores (Rhizopus): all **asexual**.
- **Sexual** reproduction combines DNA from two individuals and gives more variation. **Meiosis** halves the chromosomes in gametes; fertilisation restores the number.
- Flower: stamen (anther + filament) and pistil (stigma + style + ovary). **Pollination** (anther → stigma) then **fertilisation** (via pollen tube). Ovule → seed, ovary → fruit.
- Puberty brings sexual maturation. Male: testes, vas deferens, seminal vesicles, prostate, urethra, penis. Female: ovaries, fallopian tubes, uterus, cervix, vagina. Fertilisation in the **fallopian tube**; nourishment through the **placenta**; no fertilisation → **menstruation**.

- Contraception: condoms, oral pills, Copper-T, surgery. Condoms also reduce STD spread. Sex-selective abortion is illegal; the sex ratio must be maintained.

Quiz Test yourself

Tap an answer. **Score: 0 / 15**

1. Asexual reproduction takes place through budding in (NCERT Ex. 1)

- Amoeba
- **Yeast**
- Plasmodium
- Leishmania

Yeast puts out small buds that separate and grow.

2. Which is not a part of the female reproductive system? (NCERT Ex. 2)

- Ovary
- Uterus
- **Vas deferens**
- Fallopian tube

Vas deferens carries sperms in males.

3. The anther contains (NCERT Ex. 3)

- Sepals
- Ovules
- Pistil
- **Pollen grains**

The anther of the stamen makes pollen.

4. Binary fission in a definite orientation occurs in

- Amoeba
- **Leishmania**
- Plasmodium
- Yeast

Leishmania's whip-like structure fixes the plane of division.

5. Leishmania causes

- Malaria
- **Kala-azar**
- Sleeping sickness
- Typhoid

Plasmodium causes malaria.

6. Spirogyra reproduces by

- Budding
- **Fragmentation**
- Spores
- Multiple fission

It breaks into pieces on maturing.

7. In Rhizopus, the non-reproductive thread-like parts are

- Sporangia
- Spores
- **Hyphae**
- Buds

Sporangia hold the spores; hyphae are not reproductive.

8. A plant that has lost the capacity to produce seeds is

- Mustard
- **Banana**
- Pea
- Wheat

Banana, orange, rose, jasmine are grown vegetatively.

9. In tissue culture, the mass of rapidly dividing cells is called

- Embryo
- **Callus**
- Zygote
- Sporangium

The callus is then given hormones to form plantlets.

10. If human body cells have 46 chromosomes, a sperm has

- 46
- **23**
- 92
- 12

Meiosis halves the number.

11. After fertilisation, the ovary develops into the

- Seed
- **Fruit**
- Embryo
- Seed coat

Ovule → seed, ovary → fruit.

12. The future shoot in a seed is the

- **Plumule**
- Radicle
- Cotyledon
- Seed coat

Radicle = future root; cotyledon = food store.

13. In human females, fertilisation takes place in the

Uterus	Ovary
Fallopian tube	Cervix

Sperms meet the egg in the oviduct.

14. Which gland adds fluid to sperms?

Thyroid	**Prostate**
Pituitary	Adrenal

Prostate and seminal vesicles.

15. Syphilis is a

Viral STD	**Bacterial STD**
Fungal infection	Hormonal disorder

Gonorrhoea and syphilis are bacterial; warts and HIV-AIDS are viral.

NCERT Answers to all questions

Tap a question to see the answer. Use **Print or save as PDF** at the top to get all answers open.

In-text questions (page 114)

1. What is the importance of DNA copying in reproduction?

DNA carries the information for making proteins, which decide the **body design**. Copying DNA passes this blueprint to the offspring, so they have the same basic body design as the parent and can use the same niche. Small errors in copying also create **variations**, which help the species survive changes and are the basis of evolution.

2. Why is variation beneficial to the species but not necessarily for the individual?

If the niche changes drastically (e.g. water temperature rises), a population of identical individuals could be wiped out. If some individuals have variations (e.g. heat resistance), they survive and continue the species. So variation helps the **species** survive over time. For a given individual, a variation may be useless or even harmful in its present environment.

In-text questions (page 119)

1. How does binary fission differ from multiple fission?

In **binary fission** the parent cell divides into **two** daughter cells (Amoeba, Leishmania). In **multiple fission** it divides into **many** daughter cells at the same time (Plasmodium): the nucleus divides repeatedly before the cytoplasm separates.

2. How will an organism be benefited if it reproduces through spores?

A large number of spores are produced; they are light and spread easily; their **thick walls** protect them from drying and harsh conditions until they reach a moist surface, where they grow. So the organism multiplies fast and survives bad conditions.

3. Can you think of reasons why more complex organisms cannot give rise to new individuals through regeneration?

Complex organisms have highly specialised cells organised into tissues, organs and organ systems placed at definite positions. Most of their cells have lost the ability to multiply and change into other cell types, so a cut piece cannot rebuild the whole complex body.

4. Why is vegetative propagation practised for growing some types of plants?

(i) Plants that have lost the capacity to produce seeds (banana, orange, rose, jasmine) can be grown. (ii) Plants flower and fruit earlier than seed-grown ones. (iii) All new plants are genetically similar to the parent and keep its good characters. (iv) It is quick and easy (sugarcane, roses, grapes).

5. Why is DNA copying an essential part of the process of reproduction?

DNA is the blueprint of the body design. A new individual needs its own full set of this information, so the DNA must be copied and one copy given to each new cell along with cellular apparatus. Without DNA copying, the new cell would not have the instructions to make proteins and carry on life.

In-text questions (page 126)

1. How is the process of pollination different from fertilisation?

Pollination is the transfer of pollen grains from the anther to the stigma. **Fertilisation** is the fusion of the male germ-cell (brought by the pollen tube) with the female germ-cell in the ovule, forming a zygote. Pollination comes first and happens on the stigma; fertilisation follows inside the ovary.

2. What is the role of the seminal vesicles and the prostate gland?

They add their secretions to the sperms, so the sperms are in a **fluid** that makes their transport easier and also provides **nutrition**.

3. What are the changes seen in girls at the time of puberty?

Breast size increases and the skin of the nipples darkens; menstruation begins; thick hair grows in the armpits and genital area; thinner hair on legs and arms; the skin becomes oily and pimples may appear.

4. How does the embryo get nourishment inside the mother's body?

Through the **placenta**, a disc embedded in the uterine wall. It has villi on the embryo's side and blood spaces on the mother's side, giving a large surface area for glucose and oxygen to pass from the mother's blood to the embryo. Wastes from the embryo pass back into the mother's blood.

5. If a woman is using a copper-T, will it help in protecting her from sexually transmitted diseases?

No. A copper-T is placed in the uterus and only prevents pregnancy. It does not stop contact with body fluids, so it gives no protection against STDs. A condom (barrier) is needed for that.

Exercises (page 127)

1. Asexual reproduction takes place through budding in (a) Amoeba (b) Yeast (c) Plasmodium (d) Leishmania.

(b) Yeast.

2. Which of the following is not a part of the female reproductive system in human beings? (a) Ovary (b) Uterus (c) Vas deferens (d) Fallopian tube

(c) Vas deferens (it is male).

3. The anther contains (a) sepals (b) ovules (c) pistil (d) pollen grains.

(d) Pollen grains.

4. What are the advantages of sexual reproduction over asexual reproduction?

(i) It combines DNA from two individuals, creating **new combinations of variations** much faster. (ii) More variation helps the species survive changes in the environment. (iii) Variation is the basis of evolution of new species. (iv) Harmful effects of accumulated errors are less likely to spread to all offspring.

5. What are the functions performed by the testis in human beings?

(i) Formation of male germ-cells, the **sperms**. (ii) Secretion of the hormone **testosterone**, which regulates sperm formation and brings about the changes in boys at puberty.

6. Why does menstruation occur?

Every month the ovary releases an egg and the uterus lining becomes thick and spongy, richly supplied with blood, to nourish an embryo. If the egg is **not fertilised**, this lining is not needed, so it breaks down and comes out through the vagina as blood and mucous. This is menstruation; it lasts about 2–8 days.

7. Draw a labelled diagram of the longitudinal section of a flower.

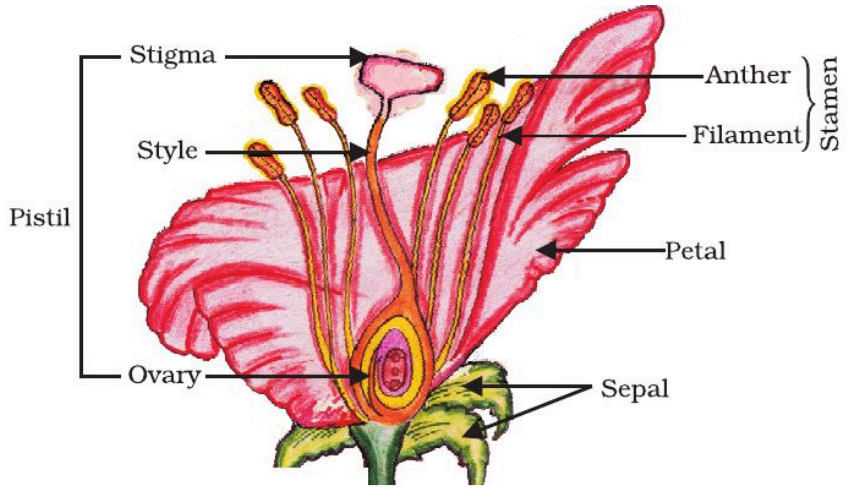


Fig. 7.7: label stigma, style, ovary (pistil); anther, filament (stamen); petal; sepal

8. What are the different methods of contraception?

(i) **Barrier methods**: condom on the penis or similar covering in the vagina. (ii) **Hormonal methods**: oral pills that stop eggs being released. (iii) **Intra-uterine devices**: loop or copper-T in the uterus. (iv) **Surgical methods**: vasectomy (vas deferens blocked) and tubectomy (fallopian tube blocked).

9. How are the modes for reproduction different in unicellular and multicellular organisms?

Unicellular organisms reproduce by simple cell division: binary fission (Amoeba, Leishmania), multiple fission (Plasmodium) or budding (yeast). **Multicellular** organisms have specialised tissues and organs, so they cannot divide cell by cell. Simple ones use fragmentation, regeneration, budding, spores or vegetative propagation; complex ones have specialised reproductive organs and cells and reproduce sexually.

10. How does reproduction help in providing stability to populations of species?

Reproduction replaces individuals that die and keeps numbers up. Because DNA copying is fairly **consistent**, offspring keep the body design that suits the species' niche. The small variations it creates help some individuals survive if the niche changes. So populations stay stable over time.

11. What could be the reasons for adopting contraceptive methods?

(i) To avoid unwanted pregnancy and protect the woman's health. (ii) To space births and plan family size. (iii) To control population growth and improve living standards. (iv) Condoms also reduce the spread of STDs like HIV-AIDS, gonorrhoea and syphilis.

Diagrams from the NCERT Science textbook for Class 10 (reprint 2026-27). Board questions from CBSE Class 10 Science papers 2023 to 2026.